

# LUCAS NEWCOMB

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## EDUCATION

### University of Maryland, College Park

*B.S. Computer Engineering, A. James Clark School of Engineering*

College Park, MD

*Expected May 2029*

### Rochester Institute of Technology

*Computer Engineering, BS/MS accelerated program; GPA: 3.98/4.00, Dean's List*

Rochester, NY

*Aug. 2025 – May 2026*

- **Coursework (A in each):** Calculus I–II, University Physics I, Computer Science I–II, Digital System Design I

## TECHNICAL EXPERIENCE

### Embedded Software Team Member

*RIT Space Exploration (SPEX) — Rover Team*

Aug. 2025 – May 2026

*Rochester, NY*

- Wrote the rover's CAN messaging library — the shared communication layer between the Teensy 4.1 controllers running the arm, chassis, and science subsystems — using dynamic dispatch to route incoming frames to registered per-message handlers, so subsystem code declares what it listens for instead of parsing the bus itself.
- Authored much of the rover's microcontroller firmware in C++ on PlatformIO; the arm and life-detection subsystems both ran on the library.
- Helped build a software simulator that mocks the rover's subsystems, letting the team exercise and test subsystem code without the physical rover.

### Student Intern

*U.S. Army Research Laboratory*

July 2024 – Aug. 2024

*Adelphi, MD*

- Developed a prototype grant proposal application on ServiceNow, building the workflow from submission through review.
- Worked within an agile software development team alongside full-time staff.

### Electrical Assistant

*Cox Farms*

Summers 2021, 2023

*Centreville, VA*

- Built and wired electrical rigs for Fields of Fear attractions; wired and installed outlets for a new barn build.

## PROJECTS

### CUBERT — Flatsat Flight Software & Ground Station | *Rust, RTIC, STM32F446, Python*

2026 – Present

- Built end-to-end flight software for a flatsat: `no_std` Rust firmware on an STM32F446 that samples sensors on a periodic schedule, sequences behavior through a state machine, and detects and recovers from faults at runtime.
- Used RTIC for deterministic, priority-based task scheduling with no dynamic allocation, and framed downlinked telemetry over serial with COBS so packet boundaries survive a noisy link.
- Wrote the matching ground station — a Python web dashboard that decodes the live telemetry stream and renders it as time-series graphs, current values, and subsystem status indicators.

### LUNA — 3D N-Body Simulator | *C++, OpenGL*

2025 – Present

- Built a real-time gravitational N-body simulator with two interchangeable solvers — a Barnes–Hut octree at  $O(n \log n)$  and direct summation at  $O(n^2)$  as an accuracy reference — integrated with a symplectic leapfrog scheme that keeps orbital energy stable over long runs instead of drifting.
- Multithreaded the force computation to sustain ~160 FPS at 10,000 bodies while remaining interactive at 100,000; renders live in OpenGL and runs headless for batch jobs.

## ADDITIONAL EXPERIENCE

### Robotics Instructor

*Unity Active*

June 2026 – Aug. 2026

*Northern Virginia*

- Taught robotics to school-age students with no prior technical background.

### Team Member, promoted to Shift Manager

*Papa John's*

Jan. 2025 – Aug. 2026

*Burke, VA*

- Worked the make line and register through high-volume shifts; promoted to shift manager in summer 2026.

## TECHNICAL SKILLS

**Languages:** C, C++, Rust, Python

**Embedded:** STM32, Teensy, CAN, I2C, SPI, UART

**Systems & Tools:** Linux, Git, CMake/Ninja, PlatformIO, OpenGL, clang-tidy, cppcheck, ServiceNow